Exercises Analytics and Knowledge Discovery WS 2025

Principal Components Analysis (PCA)

PC1 Apply the PCA to the superhero-data.
Hint: For a sample analysis in Python see `https://www.reneshbedre.com/blog/principal-component-ana` `html` Look at the iris example in the lower part.

(a) Data understanding: Briefly describe the data , i.e. what rows and columns do we have, what kind of entries are there? Based on this: should one normalize before?

(b) Calculate the PCA and give out the principal components. Show the coeffients of the first principal component.

(c) Assess the number of needed dimensions

(d) Interpretation of the transformation: show the biplot with labeled variable names and describe, what you see there: what do the first 2 PC's mean?

PC2 PCA for MNIST digit data.
Tip: choose a random subset of the data for doable computation times.

(a) Data understanding: Briefly describe the data , i.e. what rows and columns do we have, what kind of entries are there? SBased on this: should one normalize before?

(b) Calculate the PCA and assess the number of needed dimensions.

(c) Assess the quality of dimension reduction by visualizing the transformed data (by reducing the data to 2 dimensions and color coding of the 10 different digits). Can the different digits be separated in the reduced space? Is this result expectable based on the findings of question (b)?

PC3 Movie data

(a) Data understanding: Briefly describe the data , i.e. what rows and columns do we have, what kind of entries are there? Based on this: should one normalize before?

(b) Apply a PCA to the movie data. How many dimensions should we keep in order to not loose too much information?

(c) Interpretation trial: Show the biplot with labeled variable names and describe, what you see there. How useful is this plot? Why or why not (depending on your answer before)?